

Real Time Mock Interview Evaluation using CNN

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Abstract—In the wake of technological advancements and the shift in trends triggered by the COVID-19 pandemic, most traditional methods have been replaced by more convenient online alternatives. Companies nowadays assess and recruit potential candidates through interviews, which can either be one-on-one or one-to-many discussions with officials. A mock interview can help you develop interview strategies, improve your communication skills, answer difficult questions, and alleviate the nerves that many people feel before a real job interview. The main objective of the proposed system is to simplify and streamline the process of conducting mock interviews. The system is designed to be fair and impartial, eliminating the need for human intervention and reducing the effort required. The system uses various techniques such as facial recognition, speech analysis, emotion recognition, sentiment analysis, and video analysis to analyse the candidates' responses and determine their personality traits. It then generates feedback based on its analysis, providing a valuable tool for improving interview skills. The system is built on a foundation of extensive model training. CNN is implemented for facial recognition, DNN for speech and emotion recognition, OpenCV for video analysis, and Natural Language Processing for question-answer matching and feedback generation. Its main objective is to assist candidates in preparing for an interview and outperforming a regular interview

Keywords—CNN, Facial recognition, Speech Analysis, OpenCV, Natural Language Processing

I. INTRODUCTION

Mastering the art of interviews is a critical skill for career success in today's competitive job market. Mock interviews are a valuable tool for job seekers to practice and improve their interviewing skills. Traditionally, these sessions involve human evaluators who provide feedback on the candidate's performance. However, in recent years, technology has paved the way for innovative approaches to mock interview evaluations.

The research objective is to simulating a similar environment is through the use of CNN (Convolutional Neural Network) to create a comprehensive and dynamic platform for interview practice and improvement. This innovative approach to mock interview evaluation not only saves time but also ensures a more objective and data-driven assessment. It can be a valuable tool for job seekers looking to fine-tune

their interview skills and gain insights into their strengths and weaknesses.

The platform is intended to help job candidates and students improve their interview skills. The primary goal of this project is to develop an automated system that simulates interview scenarios and provides feedback based on the user's performance. The project seeks to bridge the gap between theory and practice by providing users with valuable insights and constructive feedback to improve their interview skills and increase their chances of success in the job market.

II. LITERATURE SURVEY

The system acts as an intermediary between the actual interview and preparation mode. It assesses candidates based on three factors: emotion, confidence, and knowledge base. Emotion is evaluated using a deep learning algorithm called CNN. Confidence is judged based on speech recognition using NLP and PyDub audio. For knowledge assessment, keyword mapping and semantic mapping are used. Additionally, a web scraping module extracts keywords from received replies and maps them to online resources [1].

The system uses video, speech, and text analysis to evaluate various aspects, including sentiment, head posture, voice, and intrinsic attributes, along with an aggregate interview score, to help recruiters choose potential applicants. The interview system, which focuses on analysing interviews and assists recruiters in selecting potential applicants using an automated scoring method [8]. The system combines visual, auditory, and textual elements to analyze emotions, head posture, voice, personality traits, and intrinsic attributes, along with evaluating overall interview performance [2]. The web application is designed to assist users in preparing for online interviews. It includes a speech-to-text feature that checks the grammar of the candidate's response and provides suggestions for corrections. The results are presented in a graphical format that allows users to compare two or more interviews, enabling them to track their progress over time [4].

The authors have used machine learning techniques to identify and analyze changes in interviewee behaviour and personality characteristics based on nonverbal clues such as emotions, eye movements, grins, and head movements [7].

The system is designed to conduct interviews with candidates whose resumes have been approved by the admin. The candidate's details are collected through a website, which is then viewed by the admin. If the candidate meets the requirements, a mail is sent to their registered email ID. It analyzes their personality traits through video and speech interviews and generates a report at the end [10].

The authors used EEG to track emotions and the arousal and valence model for the classification and identification of emotion. The model distributes the emotions in 2-D space and has A & V dimensions. To train classifiers and categorize arousal and positive and negative valence emotional states, machine learning techniques have been used [13].

To solve the problems faced by students during interviews, the authors developed a virtual interview training system that focuses on three essential parts: three agents with distinct types of personalities. The system has helped shy college students improve their interview performances [6]. The system aims to create an Ai powered platform to increase the user's confidence, expanding their domain knowledge. It offers a transformative learning experience, enabling users to engage in lifelike interview simulations conducted by 3D avatar. The users can also take aptitude test and receive feedback at the end [3].

The authors designed a speech communication disorder model (SCD) prediction model using machine learning to explore eye movement during facial emotion recognition tasks and show their correlation with SCA. They also used EEG and speech signals to overcome the difficulties of recognizing automatically using naturalistic human interaction (HMI). For emotion verification, the authors used their emotion database, namely a multi-modal emotion database (MED4). The research includes four baseline algorithms, including Probabilistic Linear Discriminant analysis, Multi-Layer Perception Network, Temporal Convolution Network, and Extreme Learning Machine, and evaluated them in two different environmental conditions with a particular number of people [14].

The author talks about the survey conducted on sentiment analysis and on opinion mining. Here, analysis of people sentiments, opinions, and emotions from their text language by which we can try to understand the person's state of mood or emotion while writing the message [11].

III. PROPOSED SYSTEM

The objective of research is to provide a user-friendly and easy-to-use platform comprising:

A. Login/Sign Up

Develop a user-friendly web interface to interact with the system. Web development will be used for creating GUI for the user. This will allow user to get logged in to take up interviews or sign in for the first time as a new user. This will

enable the candidate to get first-hand information about the system, access mails, etc.

B. User Registration and Profile

Allow users to create profiles, including their name, contact information, and professional details. This stores user data securely. This lets us personalize user experience and understand on how to tailor the content for them.

C. Job Role

Provide an option for users to choose the type of interview they want to take up according their job role. This will give them an additional insight onto what things are expected of them in that particular job role.

D. AI Evaluation

1) Speech Recognition Automatic Speech Recognition (ASR):

ASR software or libraries are essential for converting spoken language into text. DeepSpeech library of the python aided by TensorFlow will be used for speech to text. It also gives us the space to create voice commands and virtual assistants.

2) Video Recognition:

Computer Vision Libraries: Open CV (Open-Source Computer Vision Library) and CNN are widely used open-source library for computer vision tasks. This lets us record the candidate's movements and analyze their emotions and decisions taken.

3) Facial Recognition:

For analysing facial expressions, libraries of deep learning frameworks like TensorFlow and PyTorch with pre-trained models will be used.

4) Gesture Recognition:

Libraries like MediaPipe by Google can help recognize hand gestures and movements to check what the candidate's body language is at that particular moment.

5) Sentiment Analysis:

Mostly NLTK library of Natural Language Processing or Hugging Face Transformers is used but there are pre-existing models BERT or GPT which can also be used to work on large datasets.

6) Answer Evaluation:

To check whether a given answer is correct or not, Natural Language Processing (NLP): NLP libraries like spaCy can help with language processing, understanding context, and extracting meaning from text.

7) Feedback and Analysis:

Store user performance data and provide analytics over time, showing improvements and areas that need more attention.

To offer detailed feedback on strengths and weaknesses, along with specific areas to focus on for improvement. Shows summary of the user's performance during the interview.

IV. METHODOLOGY

The system aims to generate constructive feedback at the end of mock interview. The flow of interview can be divided into four phases- User Registration, Select an Interview, Evaluation and Feedback as shown in Fig.1.

A. User Registration

1) Design User Interface(UI):

Develop a user-friendly registration form that enables candidates to input their details, which included their name, email address, password, and any other information necessary for registration.

Consider implementing form validation to ensure that the entered information is valid and meets your requirements.

2) Backend Implementation

Develop backend logic to handle the registration process. Create secure database tables or data structures to store candidate information.

Implement server-side validation to sanitize and validate user input before storing it in the database.

Hash and securely store passwords using industry-standard hashing algorithms (e.g., bcrypt).

3) User Authentication

Implement user authentication mechanisms to verify the identity of registered candidates.

Design a login functionality where candidates can securely authenticate using their registered email address and password.

B. Select an Interview

The candidate can choose the type of interview between technical and communication based from the selection menu. The technical interview category further allows users to select specific subjects such as Java, Data Structures and Algorithms (DSA), and Signal Processing. Each category has its own set of questions and evaluation criteria tailored to the chosen subject area.

C. Evaluation

For technical interviews, which include Java, DSA, and Signal Processing, the system checks the correctness of the candidate's answers. This involves automated code evaluation for programming-related questions and analysis of responses for theoretical questions. Utilization of techniques such as test cases for code evaluation and natural language processing for analyzing text-based answers. For the communication round, natural language processing (NLP) techniques are used to analyze the candidate's language skills, including grammar, vocabulary, fluency, etc.

D. Feedback

When a user takes up two or more mock interviews, a comparison between all the interviews is displayed. After completing an interview, the candidate's performance graph is displayed using data visualization techniques, along with their score. This graph shows the strengths and weaknesses of the candidate, and it can be used to compare different interviews taken by the candidate.

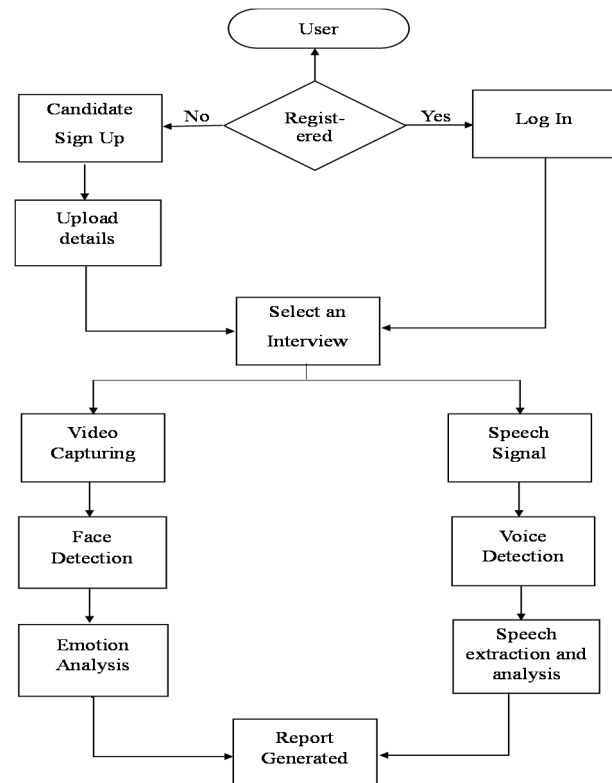


Fig. 1. Architecture of the system

V. EVALUATION RESULTS

A. EMOTION RECOGNITION

The system first connects to the candidate's camera. It detects the face and then analyses the emotion and gestures of the candidate during the interview. In Fig. 2. The candidate's emotion is recognised and used for evaluation.

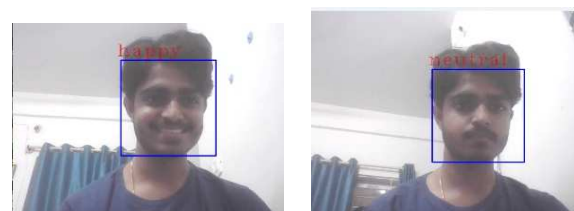
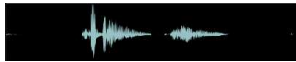


Fig. 2. Candidate emotion detected by the system

B. SPEECH RECOGNITION

In this module, the candidate's audio is converted into text. The system connects to the candidate's microphone. In Table 1, displays the input recorded by the system and the converted text as output.

TABLE I. AUDIO INPUT

Speaker	Input	Output
1.		Good morning! How are you?
2.		I am currently studying in NHCE
3.		I have always had keen interest in the development sector in IT field
4.		Walk me through your resume
5.		I had interned at XYZ company as a web designer for a period of three months.

VI. CONCLUSION

In conclusion, the CNN-driven mock interview evaluation system signifies a technological leap in interview preparation. Beyond the reliance on human evaluators, the data-driven method provides an efficient and objective alternative. This simulated interview environment facilitates skill enhancement, offering candidates valuable performance insights. Embracing technological advancements, the approach fosters effective and personalized preparation for real-world interviews. Tailored for students in technical and communication rounds. The integration of CNN technology not only delivers real-time experience but also ensures automated feedback based on user performance, streamlining and enhancing candidate preparation. Through the valuable insights gained and the practical feedback received can enhance their chances of success in the job market. Navigating the intersection of technology and professional development, the open-source project emerges as a valuable resource for students, providing comprehensive support on their journey to interview success. The system records the interaction and presents it to the user, improving the learning experience. Overall, this system is a valuable tool for job seekers looking to fine-tune their interview skills.

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